



Hands-On Data Science & Machine Learning: From Fundamentals to Cutting-Edge Generative AI Models

Training Agenda

Date: Saturday 07.12.2024

Location: ARAB AMERICAN UNIVERSITY - lab Eng203

Part 1: Foundation of Machine Learning and Data Science		6 hour
09:00 - 09:30	Opening and Training Course Overview	
09:30 - 11:00	Introduction and Fundamentals of ML	
11:00 - 11:30	Coffee Break	
11:30 - 12:00	ML Programming Tools and Support Software	
12:00 - 1:00	Datasets Collections & Data Representation & Data Understanding	
1:00 - 2:00	Data Preprocessing	
2:00 - 2:30	lunch	
2:30 - 3:00	Data Preprocessing	

Task1: (8/12/2024-13/12/2024)		2 hours
Each participant needs to perform:	Downloads a selected dataset for real application problem from the datastores that we used in training - Why the problem is important - Identify the results impact - The stakeholders - Understated the Nature of Data and understated the attributes	
	Apply all the learned techniques to represent and understand the data for ML application - Statistics - Visualization - Transfer the dataset to numerical values	
	Apply all the preprocessing techniques on the dataset - Data Integration - Data Cleaning - Transformation - Data redaction and Feature Selection	

Online Lecture: 9 /12/2024 Time: 18:15-19:15 via ZOOM		1 hour
Each participant	Will be held to answer the questions regarding the training on 7/12/2024 and any query about task 1.	

Date: Saturday 14.12.2024

Location: ARAB AMERICAN UNIVERSITY - lab Eng203

Part 2: Machine Learning Algorithms and detests applications		Zoom -- 4 hours
09:00 - 10:00	Classification /Prediction and Evaluation Methods	
10:00 - 11:00	Linear Regression & Logistic Regression	
11:00 - 11: 30	Break	
11:30 - 1:00	Machine learning algorithms theory and applications (Decision Trees and Random Forests, Support Vector Machines)	

Task2: (14/12/2024-24/12/2024)		2 hours
Each participant needs to:	• For the selected dataset in task 1: • Apply Linear Regression & Logistic Regression • Apply all the learned techniques of ML (DT, RF, SVM, KNN, Naïve Bayes, Ensemble methods, ect. • Evaluate the result using all evaluation methods • Understanding and explaining all type of results • Comparing the result quality and determine the best ML model	

Date: Saturday 21.12.2024

Location: ARAB AMERICAN UNIVERSITY - lab Eng203

Part 2: Machine Learning Algorithms and detests applications		6 hours
09:00 - 10:00	Machine learning algorithms theory and applications (Support Vector Machines, Naïve Bayes Ensemble methods: Bagging and Boosting)	
10:00 - 11:00	Machine learning algorithms theory and applications (Support Vector Machines, Naïve Bayes Ensemble methods: Bagging and Boosting)	
11:00 - 11:30	Machine learning algorithms theory and applications (Support Vector Machines, Naïve Bayes Ensemble methods: Bagging and Boosting)	
11:30 - 12:00	Coffee Break	
12:00 -1:00	Examples of application of the ML algorithms	
1:00-2:00	Machine learning algorithms theory and applications (Support Vector Machines, Naïve Bayes Ensemble methods: Bagging and Boosting)	
2:00 - 2:30	Snacks	
2:30 - 3:00	Examples of application of the ML algorithms	

Task2: (14/12/2024-24/12/2024)		2 hours
Each participant needs to:	• For the selected dataset in task 1: • Apply Linear Regression & Logistic Regression • Apply all the learned techniques of ML (DT, RF, SVM, KNN, Naïve Bayes, Ensemble methods, ect. • Evaluate the result using all evaluation methods • Understanding and explaining all type of results • Comparing the result quality and determine the best ML model • Apply GAs on selected datasets, Understanding and explaining all type of results	

Online Lecture: 23/12/2024 Time: 18:15-19:15 via ZOOM		1 hour
Each participant	Will be held to answer the questions regarding the training on 26/12/2024 and any query about task 3.	

Date: Saturday 28.12.2024

Location: ARAB AMERICAN UNIVERSITY - lab Eng203

Part 3: Machine Learning Algorithms and detests applications		6 hours
09:00 - 10:00	Neural Networks	
10:00 - 11:00	Neural Networks	
11:00 - 11: 30	Examples of application of the NNs algorithms	
11:30 -12:00	Coffee Break	
12:00 -1:00	Basics of image processing and filtering	
1:00-2:00	Deep Learning Models: Convolutional Neural Networks (CNNs) Examples application	
2:00 - 2:30	Snacks	
2:30 - 3:00	Deep Learning Models: Convolutional Neural Networks (CNNs) Examples application	

Task4: (28/12/2024-4/1/2025)		2 hours
Each participant needs to:	• Apply NNs to the processed dataset in task 1& 2 • Comparing the result quality between NNs and ML models • Understanding and explaining all type of results • Apply image processing and filtering on image dataset • Apply Deep Learning Models in image dataset	

Date: Saturday 04.01.2025

Location: ARAB AMERICAN UNIVERSITY – Eng 212

Part 3: Generative AI Models		6 hours
09:00 - 09:30	Fundamentals of Natural Language Processing and Language Models	
09:30 - 11:00	Generative vs. Discriminative Models	
11:00 - 11:30	Types of Generative Models	
11:30 - 12:00	Coffee Break	
12:00 - 1:00	Practical Examples & APIs	
1:00 - 2:00	Text Generation and Chatbots	
2:00 - 2:30	lunch	
2:30 - 3:00	Image Synthesis\Manipulation and Music Composition	

Task4: (05/01/2025-10/01/2025)		3 hours
Each participant needs to perform:	Use the auto-generative language model AraT5 to fine-tune for Arabic-English translation, you have to download a ready-made dataset for the machine translation and use the model from this source: https://huggingface.co/UBC-NLP/AraT5-base • Prepare the data • Download the model • Setup the code and environment • Show the results on the test split	
	Train AraBERT encoder language model on controversial loss learning learning to achieve word sense disambiguation task on any dataset available in any language • Extract dataset and transform into positive and negative examples • Access the model from this repository https://huggingface.co/aubmindlab/bert-base-arabert • Show-up the training and testing results	
	Employee ready-made GANs model to generate images based on user prompts • Download the model • Use it on Colab https://colab.research.google.com/ • Try different prompts and show up the diversity	

Online Lecture: 07/01/2025 Time: 15:00-17:00 via ZOOM		2 hours
Each participant	In this session I will introduce further language models and APIs for auto-regressive multi-modal models. Additionally, we will review the task details.	

